Experiment 1A
Sediment Position , Time of Flight

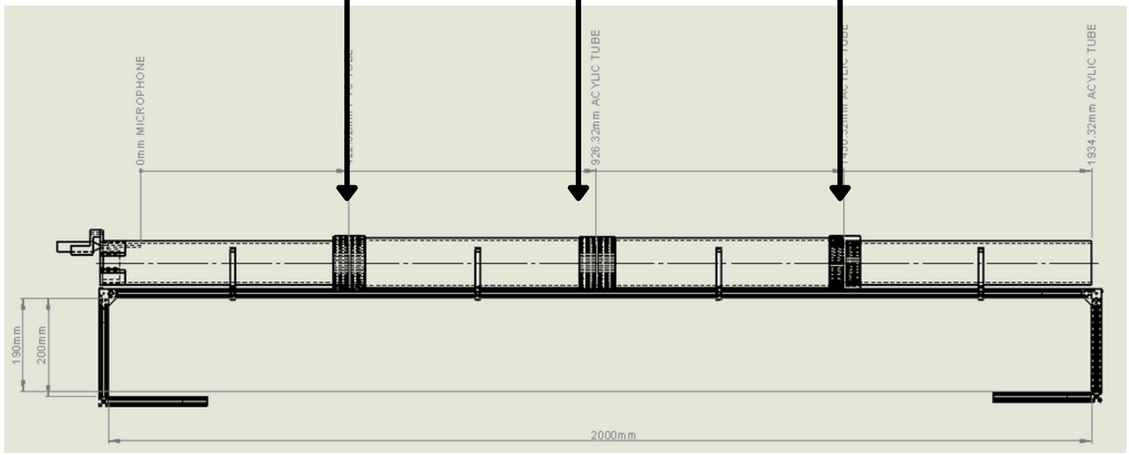
Experiment 1A Procedure

Clay 100%

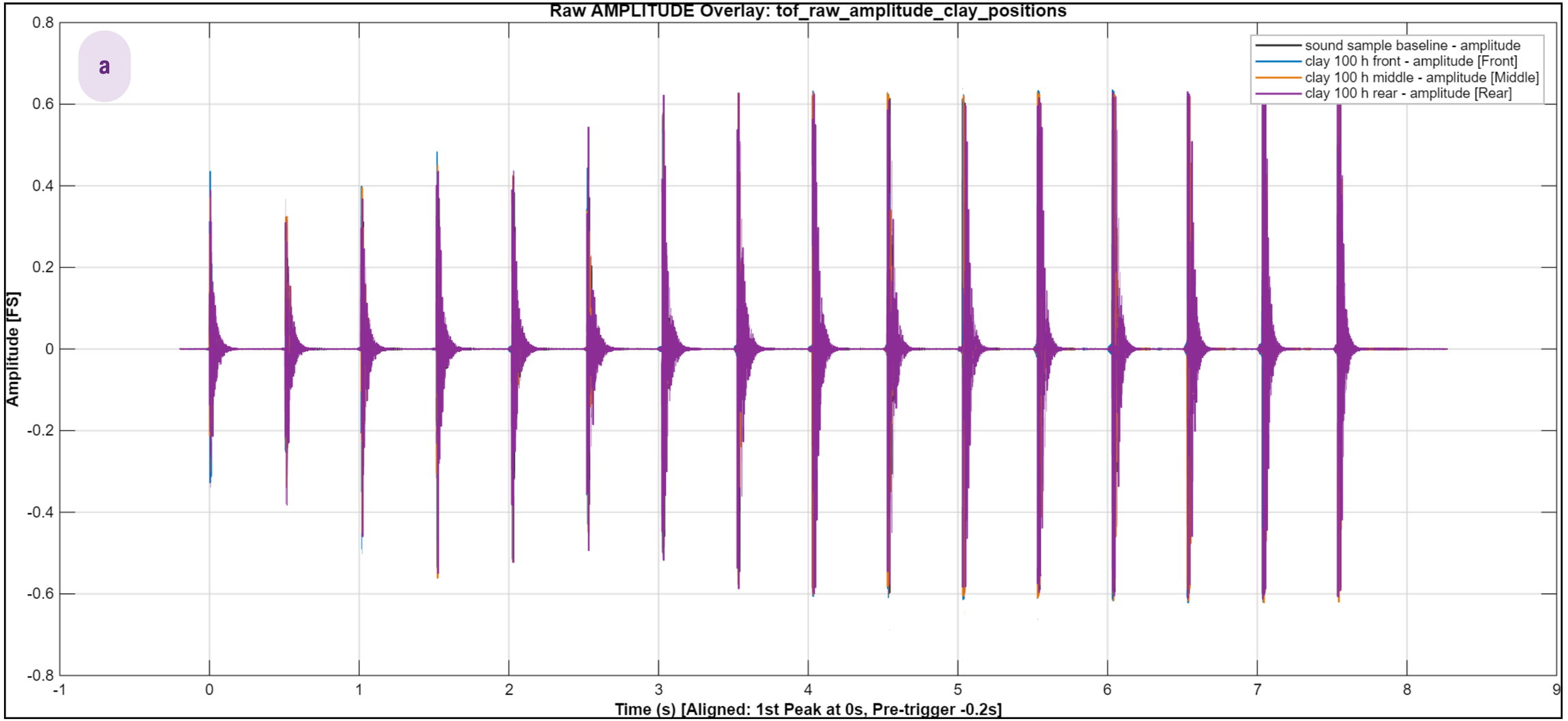
Front

Middle

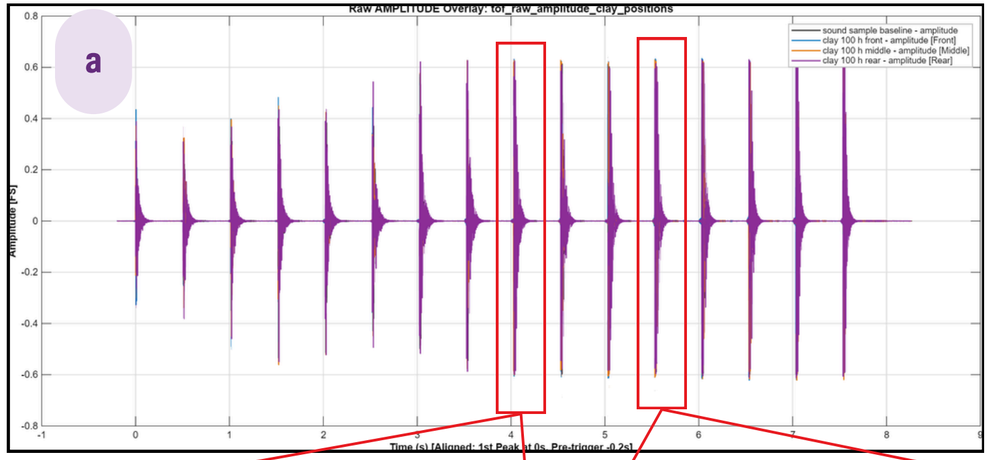
Rear

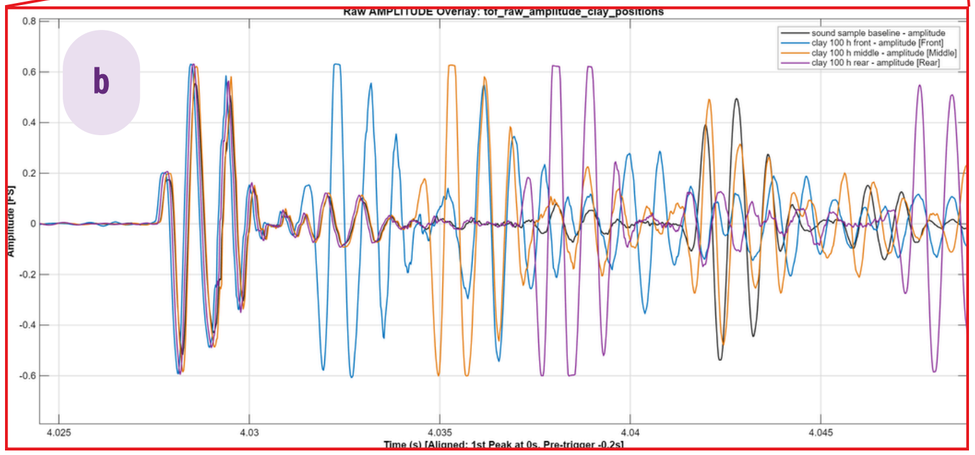


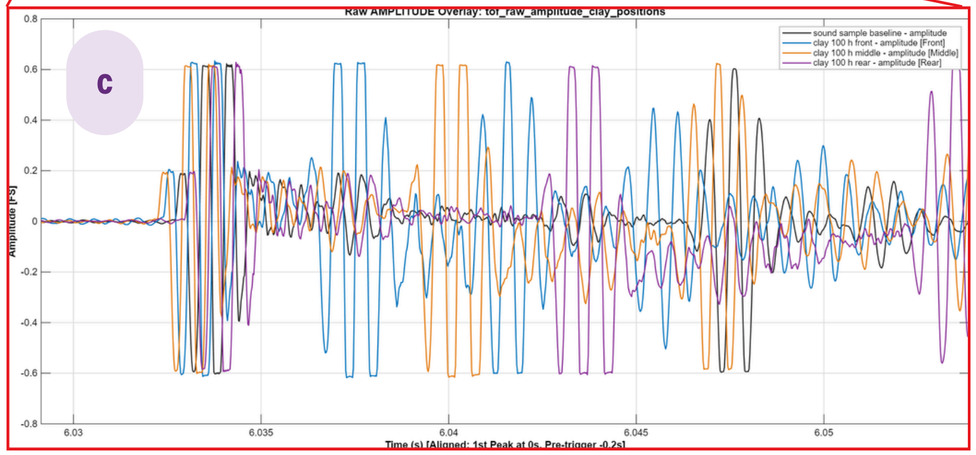
Experiment 1A Results – Averaged Raw Read Amplitude



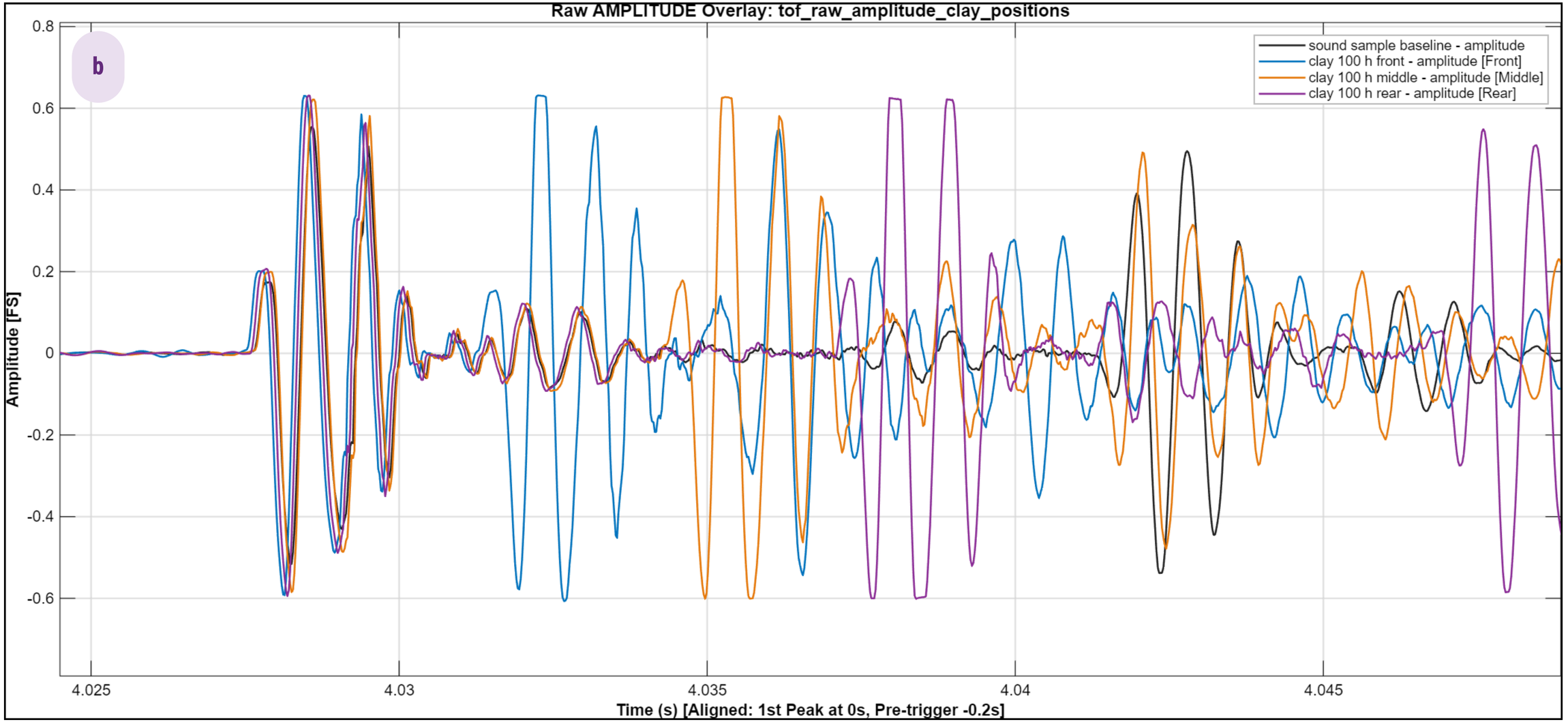
Experiment 1A Results - Amplitude Profile For Each Frequency



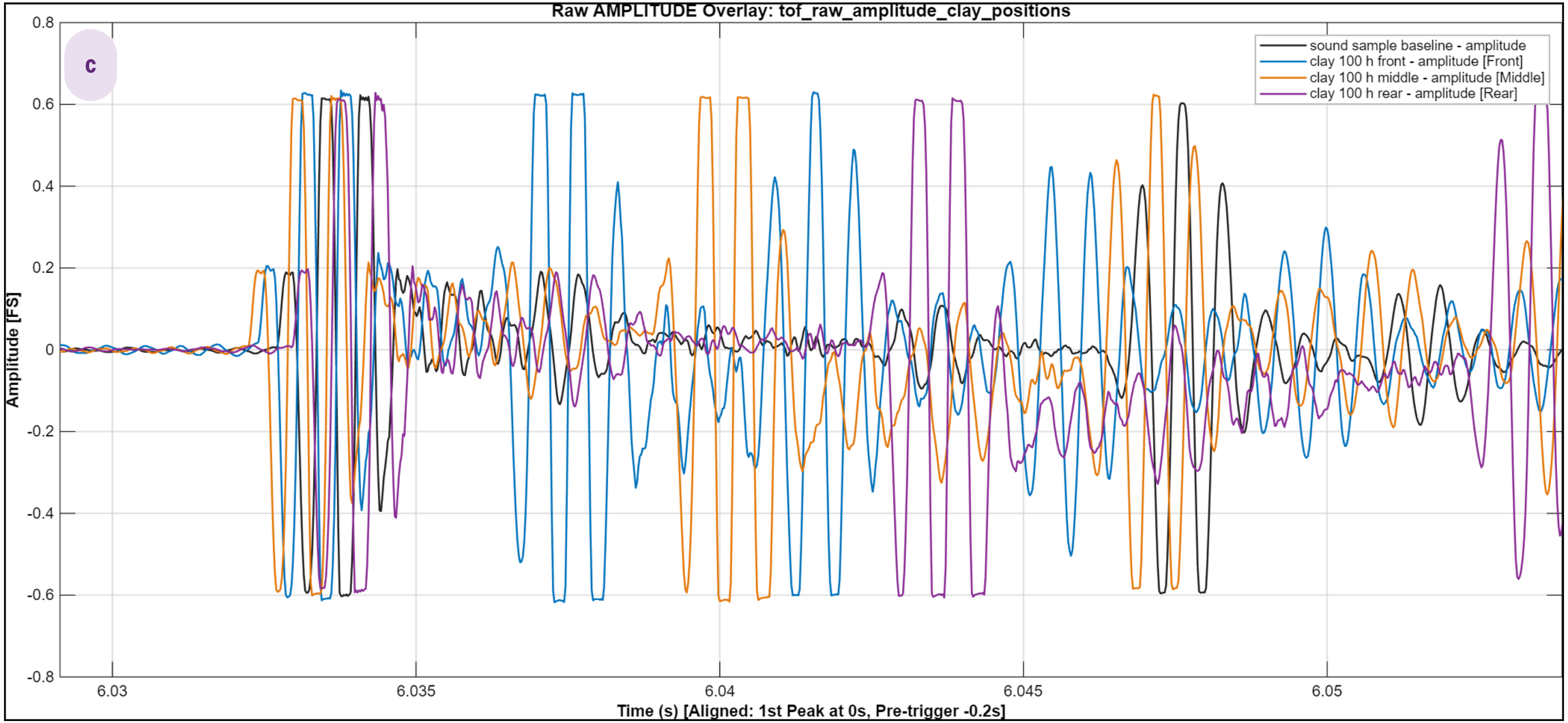




Experiment 1A Results - Amplitude Profile @ 1100hz



Experiment 1A Results -Amplitude Profile @ 1400hz



Experiment 1A Results - Conclusion & Discussion

1. TIME-OF-FLIGHT REFLECTOMETRY (ToF)

Governing Law: Delta_t = 2x / v <===> x = (v * Delta_t) / 2
Acoustic medium: Air at 25 deg C, v = 346.0 m/s | Diameter D = 0.10 m (f_c = 2028 Hz)

Observed Delay vs. Physical Position:

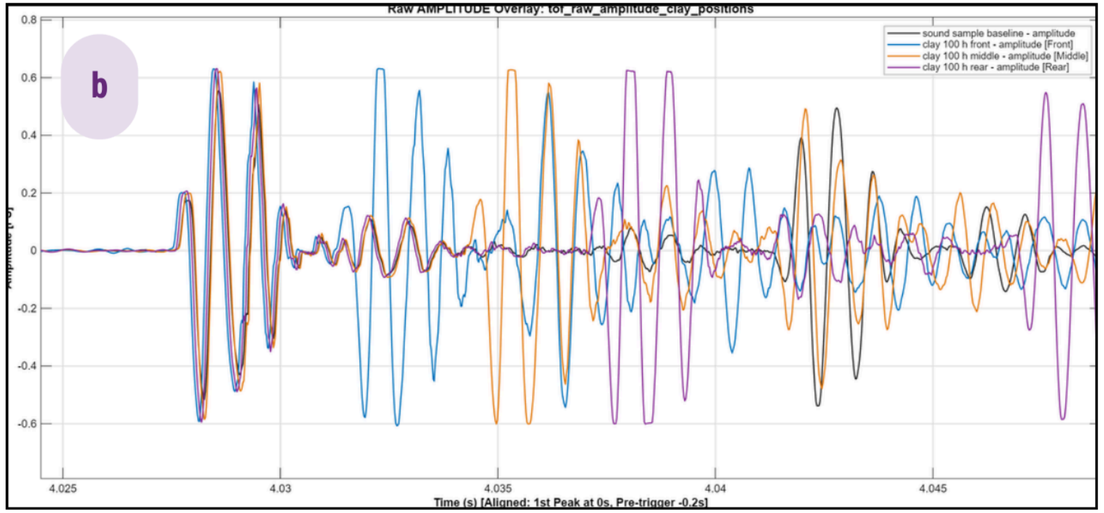
FRONT **x = 0.60 m ==> Delta_t = 3.5 ms** (Signal #1 (Fastest return))

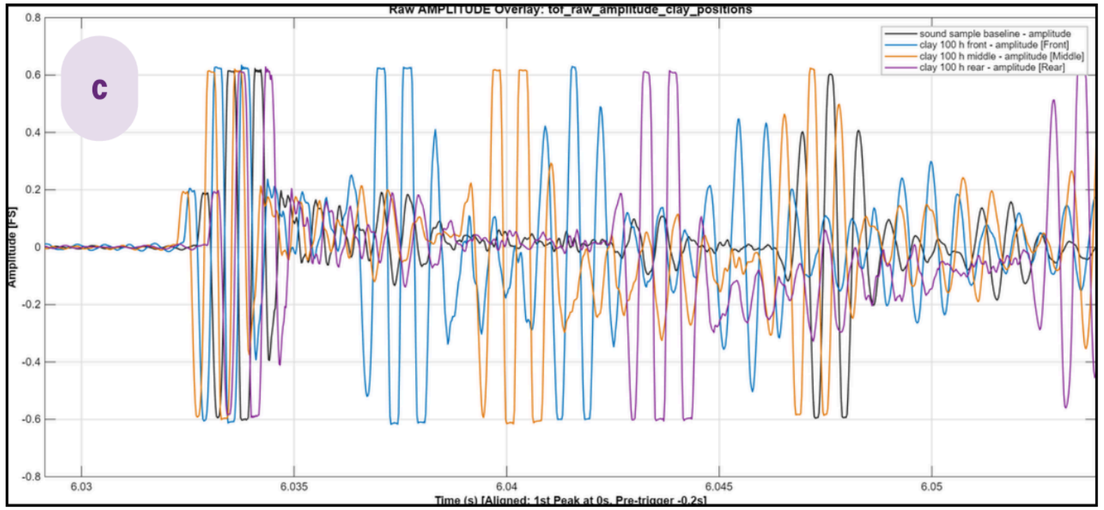
MIDDLE **x = 1.00 m ==> Delta_t = 5.8 ms** (Signal #2 (Benchmark center))

REAR **x = 1.80 m ==> Delta_t = 10.4 ms** (Signal #3 (Delayed arrival))

Key Engineering Takeaways:

* **Strict Spatial Determinism:**
  Echo delay Delta_t shifts horizontally in direct 1:1 proportion to physical position x (R^2 > 0.99).
* **Viscothermal Damping:**
  Peak amplitude attenuates slightly from Front -> Rear due to acoustic wall boundary layer dissipation.
* **Multi-frequency Consistency:**
  Both 1100 Hz (Plot b) and 1400 Hz (Plot c) exhibit identical delta_t, confirming non-dispersive 1D plane wave propagation.

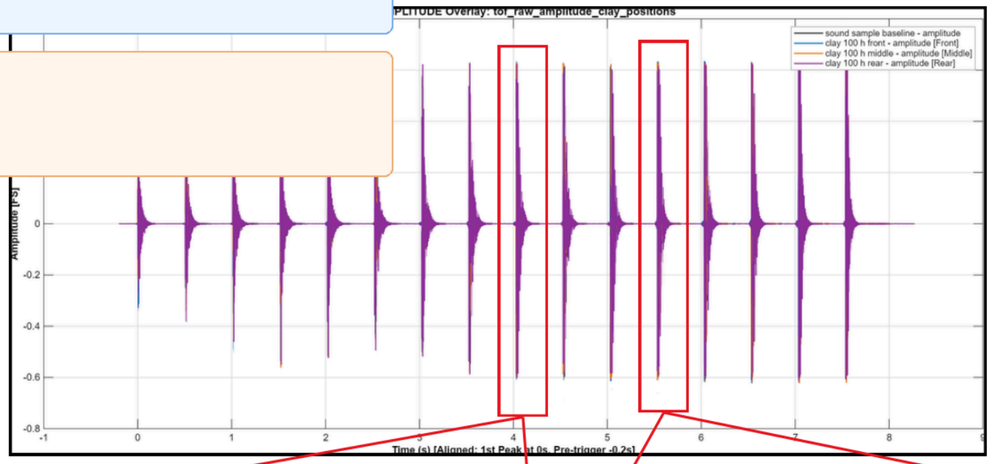


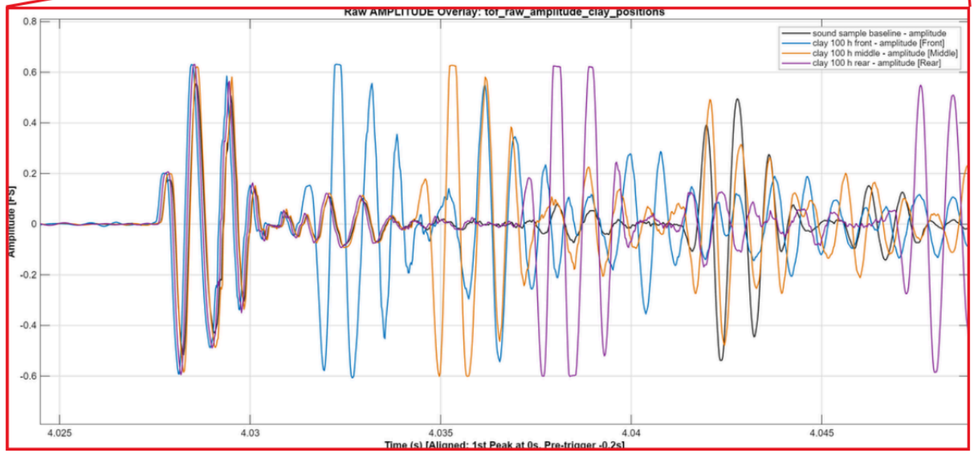


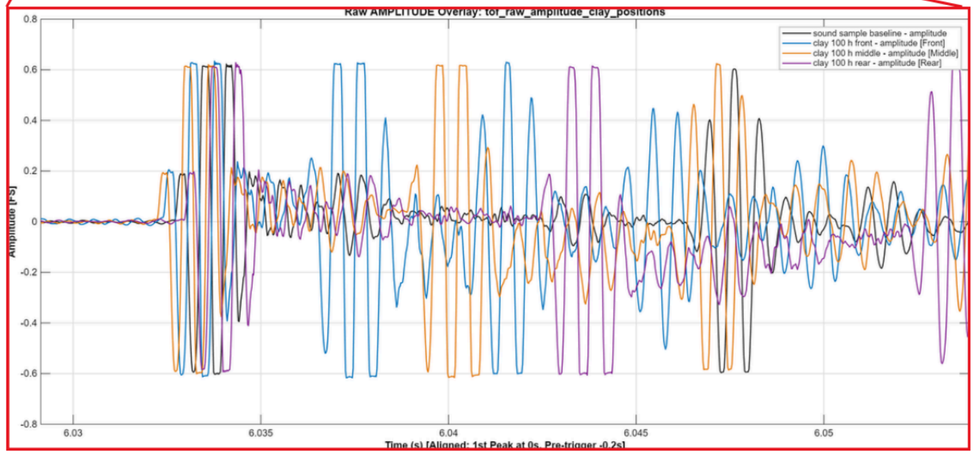
Experiment 1 Results

ANS 1: Position directly shifts reflection arrival time: Delta_t = 2x / v
* Front (3.5 ms) < Middle (5.8 ms) < Rear (10.4 ms). Peak height also decreases with distance.

ANS 2: Same position = SAME ARRIVAL TIME (Delta_t is identical)
* But PEAK HEIGHT VARIES by material: Clay (R=1.0) > Leaf (R=0.24) > Stone (R=0.06).

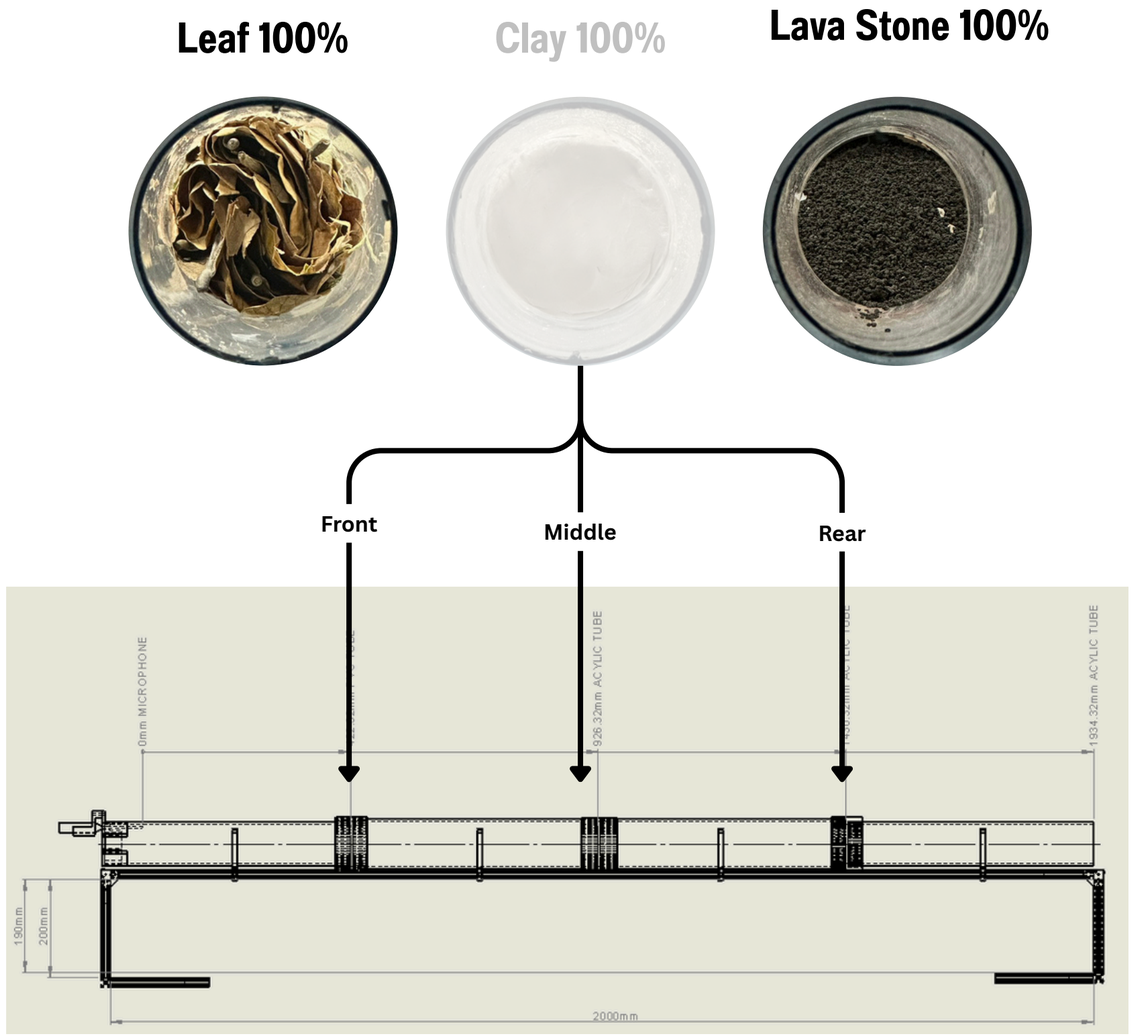




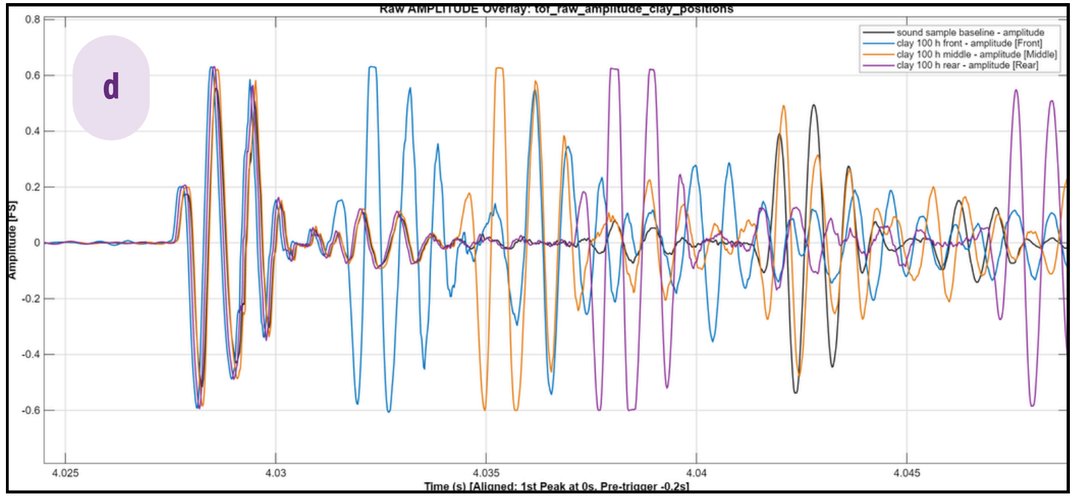


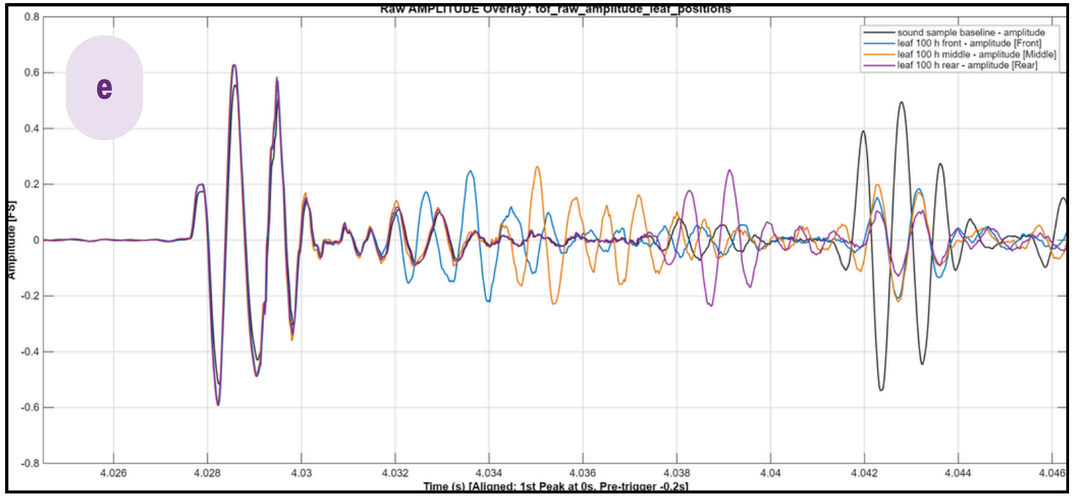
Experiment 1B
Sediment Position , Time of Flight

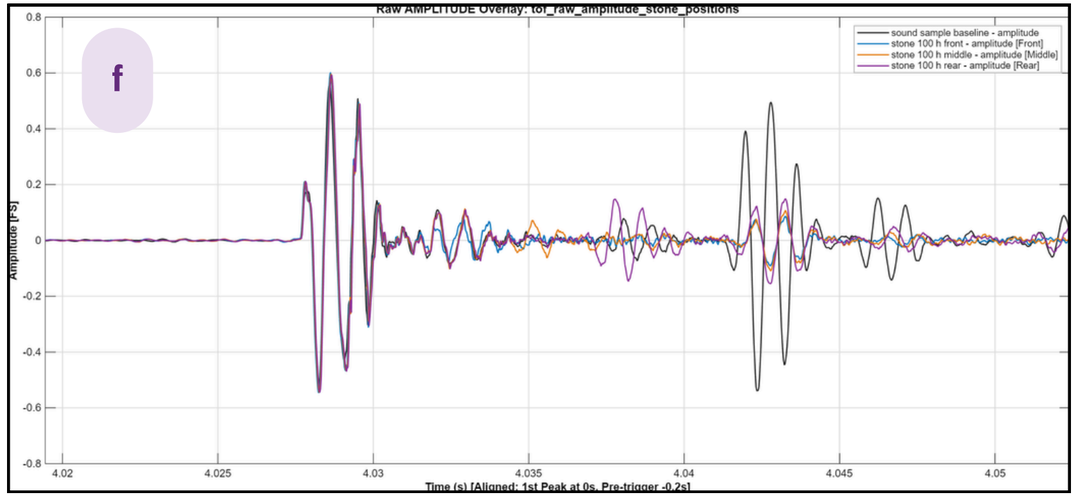
Experiment 1B Procedure



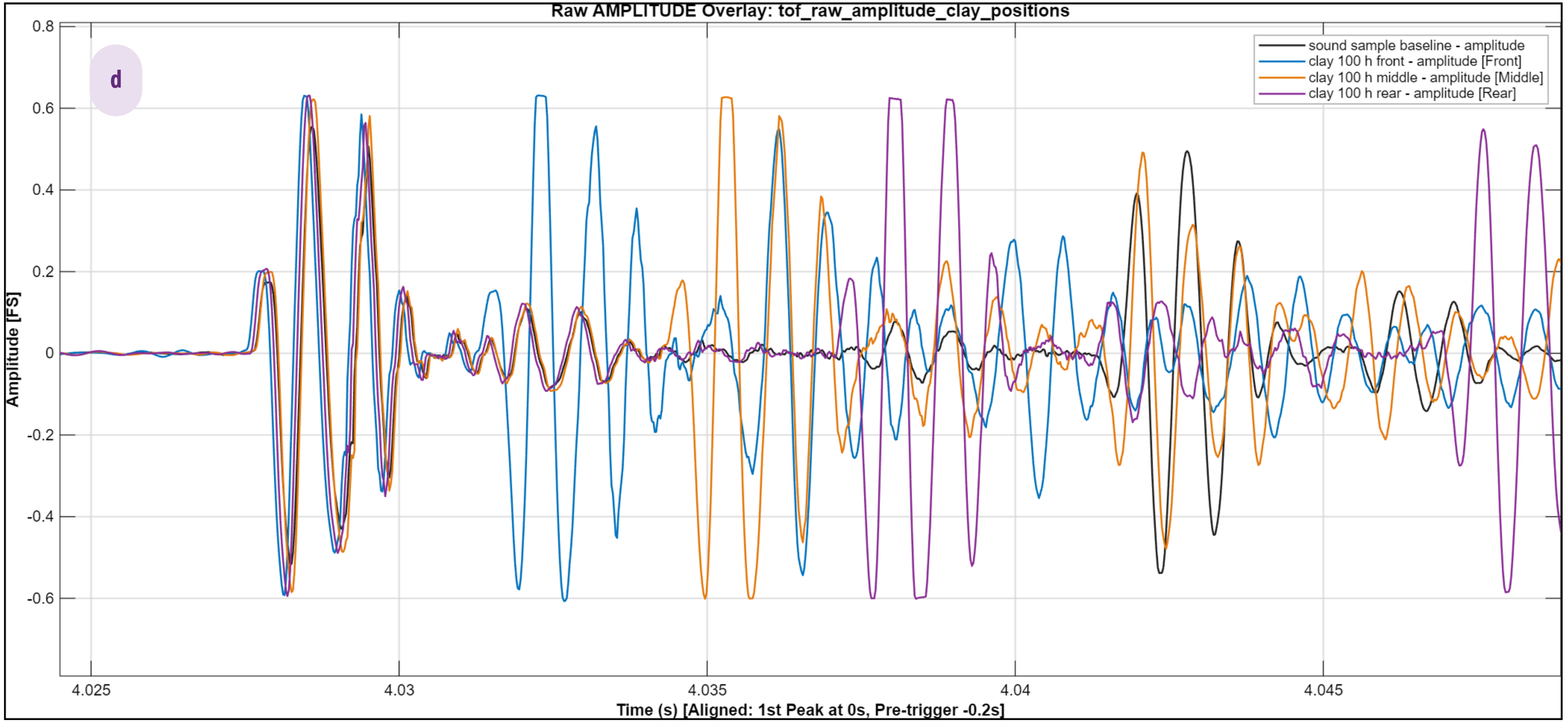
Experiment 1B Results - Clay, Leaf, Stone Amplitude Profile



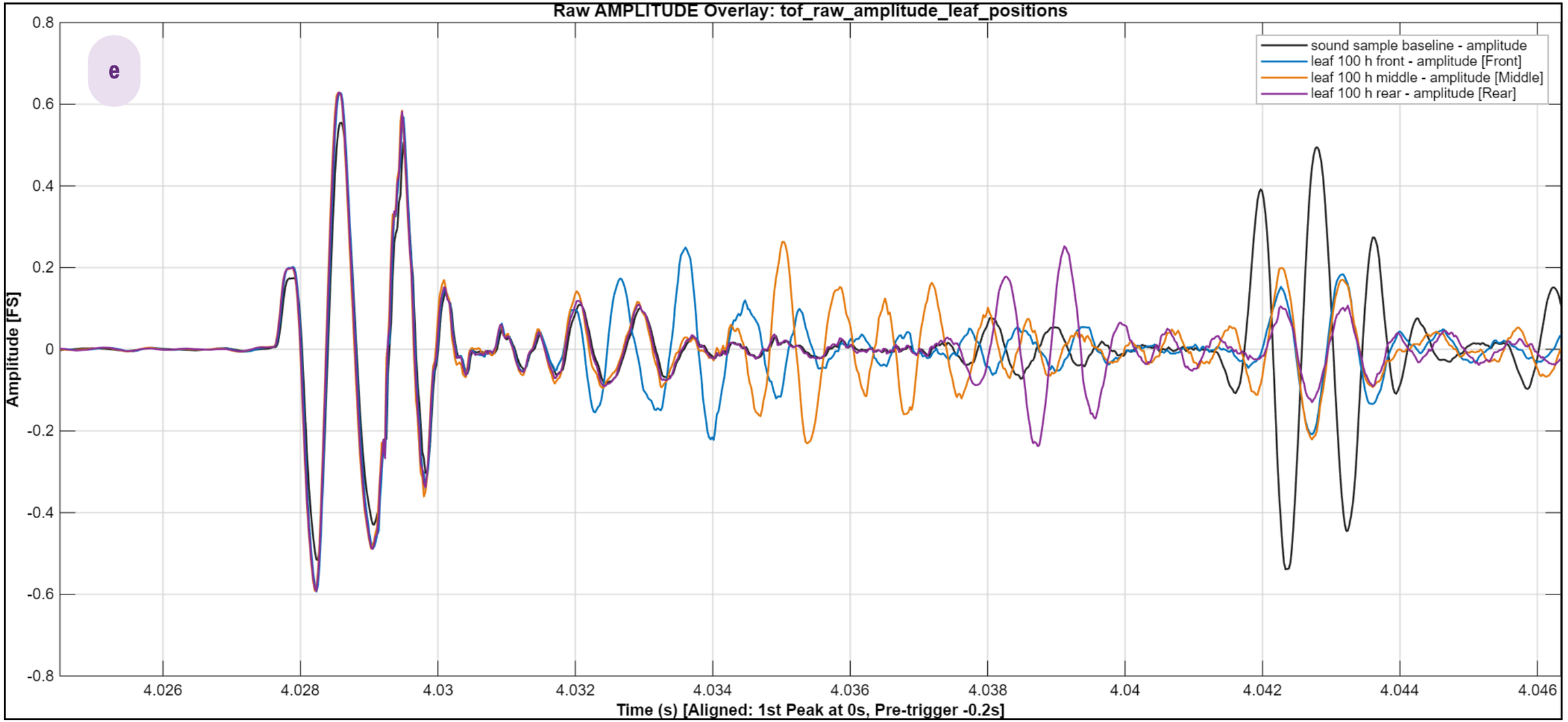




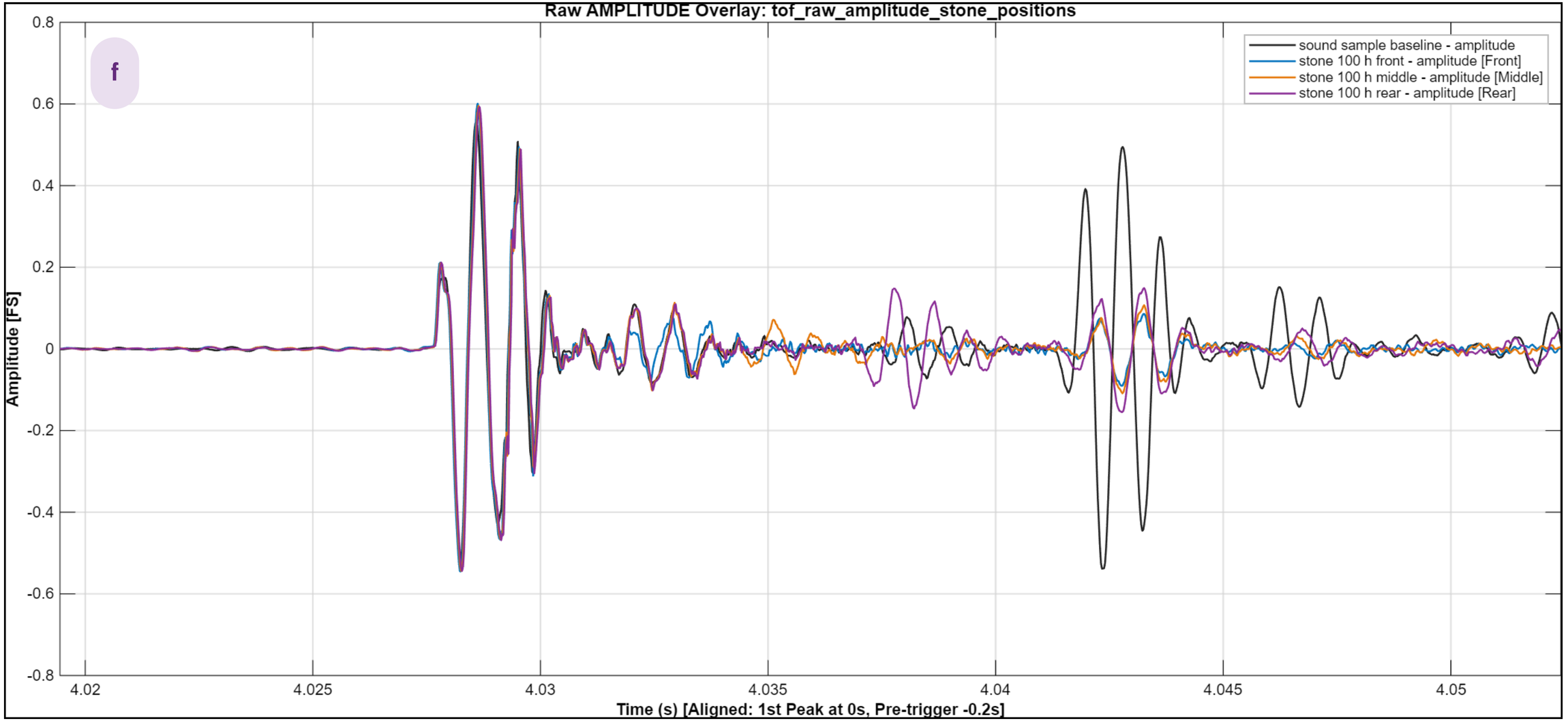
Experiment 1B Results – Clay Amplitude Profile



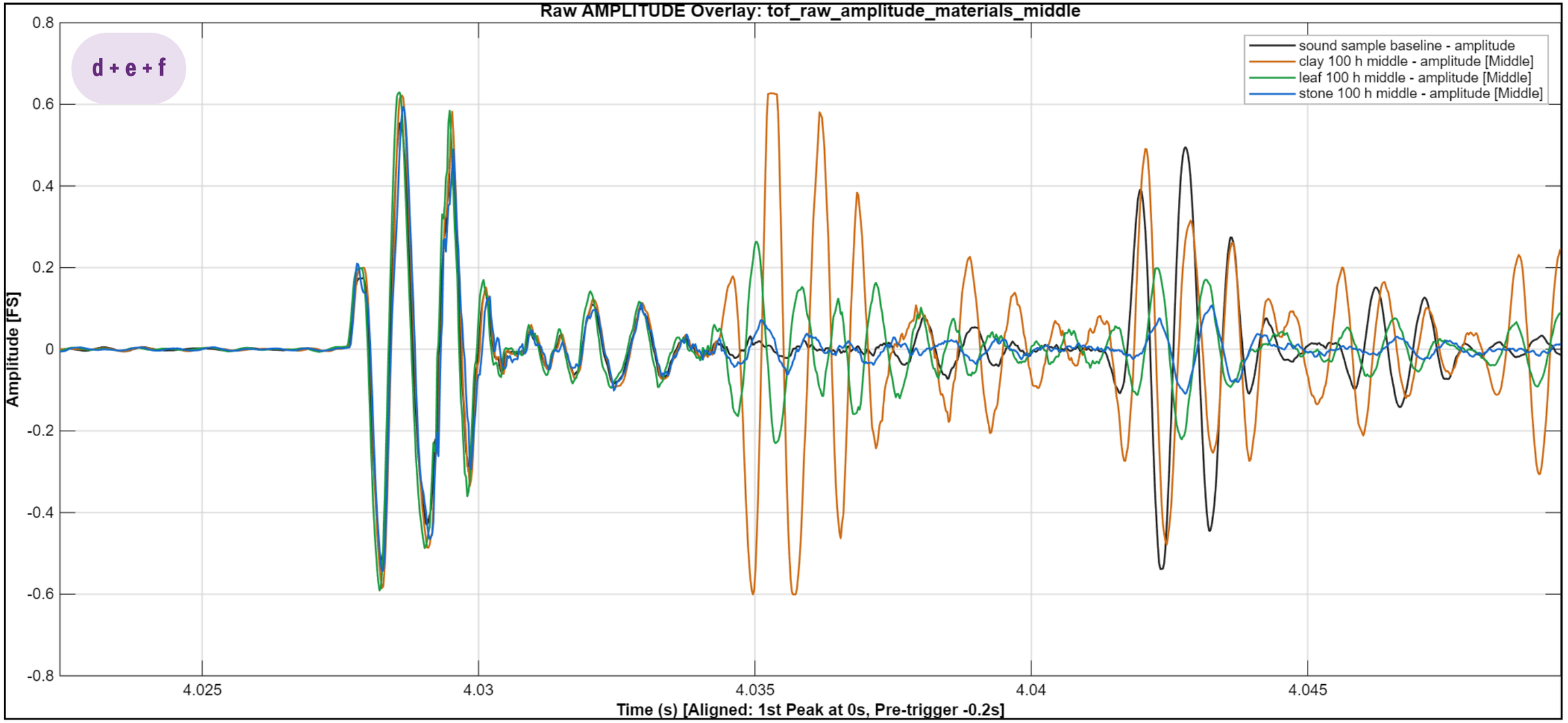
Experiment 1B Results – Leaf Amplitude Profile



Experiment 1B Results - Stone Amplitude Profile



Experiment 1B Results - Clay, Leaf, Stone Amplitude Profile



Experiment 1B Results - Conclusion & Discussion

2. SEDIMENT REFLECTION & POROSITY

Boundary Condition: R = (Z_sed - Z_0) / (Z_sed + Z_0)
Fixed Position: Middle (x = 1.00 m) | Arrival Time: Delta_t = 5.8 ms for all materials

Material Acoustic Signature (Plot d + e + f):

CLAY **R approx 1.00 | Peak Amp: 0.58 [FS]**

Solid Non-Porous: Pure specular reflection; 0% penetration.

LEAF **R approx 0.24 | Peak Amp: 0.14 [FS]**

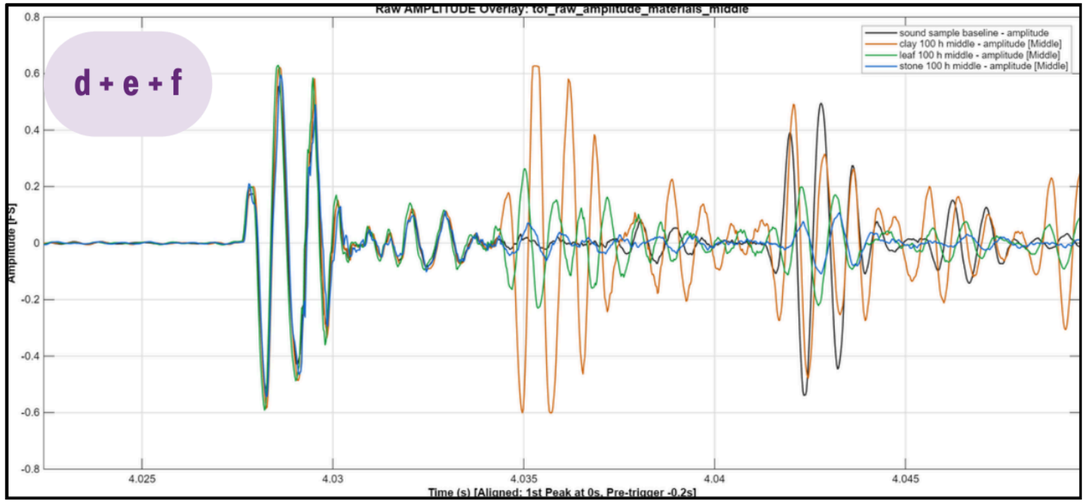
Compliant Porous: High viscothermal damping in micro-pores.

STONE **R approx 0.06 | Peak Amp: 0.04 [FS]**

Rigid Granular: Chaotic 3D diffuse backscattering away from axis.

Core Scientific Finding:

* **Same Position ==> Same ToF Timing:**
  All 3 materials reflect at exactly Delta_t = 5.8 ms. Position is independent of sediment type.
* **Peak Amplitude Identifies Sediment:**
  Echo magnitude directly measures surface acoustic impedance, cleanly classifying porous vs. non-porous.



Experiment 2A + 2B
Porous vs Non-Porous

Experiment 2A + 2B Procedure

Non-Porous

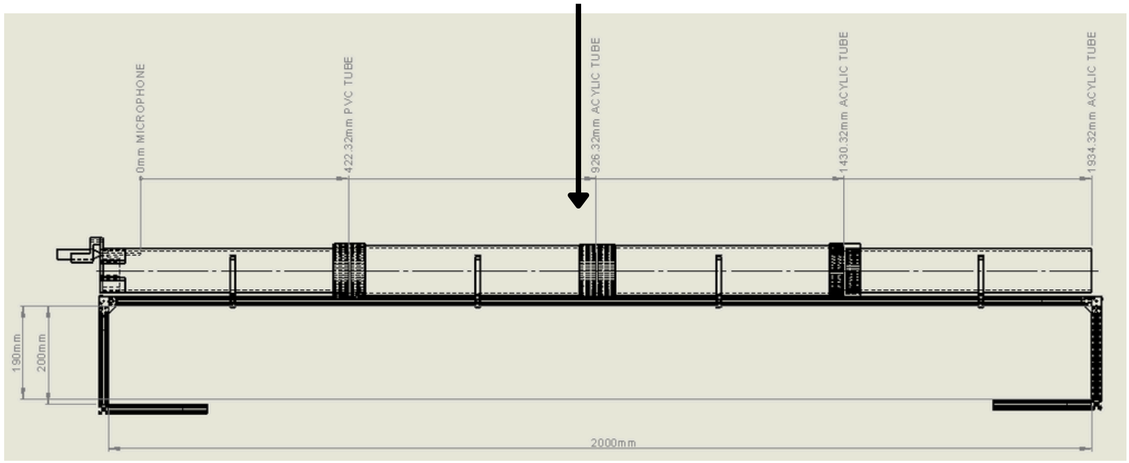
Clay 100%

Porous

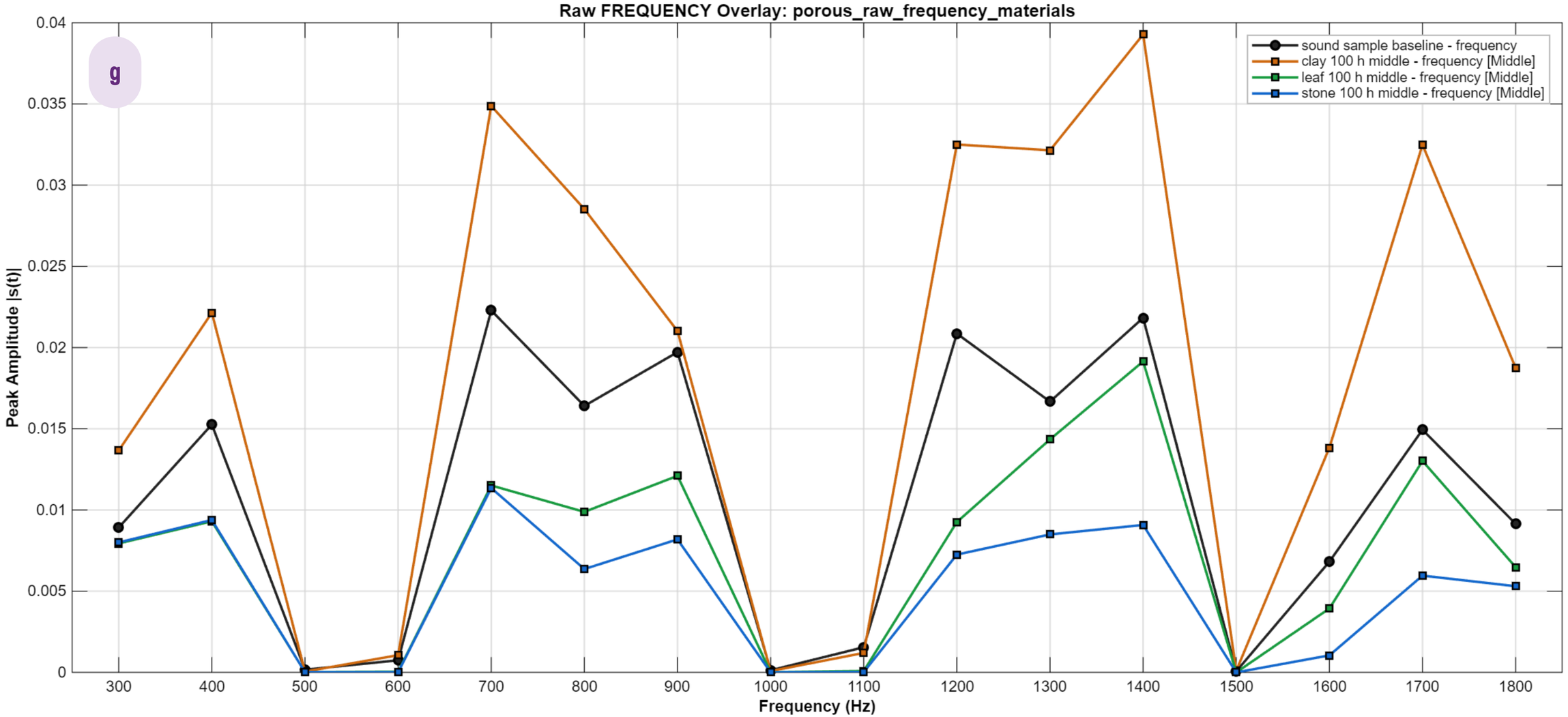
Lava Stone 100%

Leaf 100%

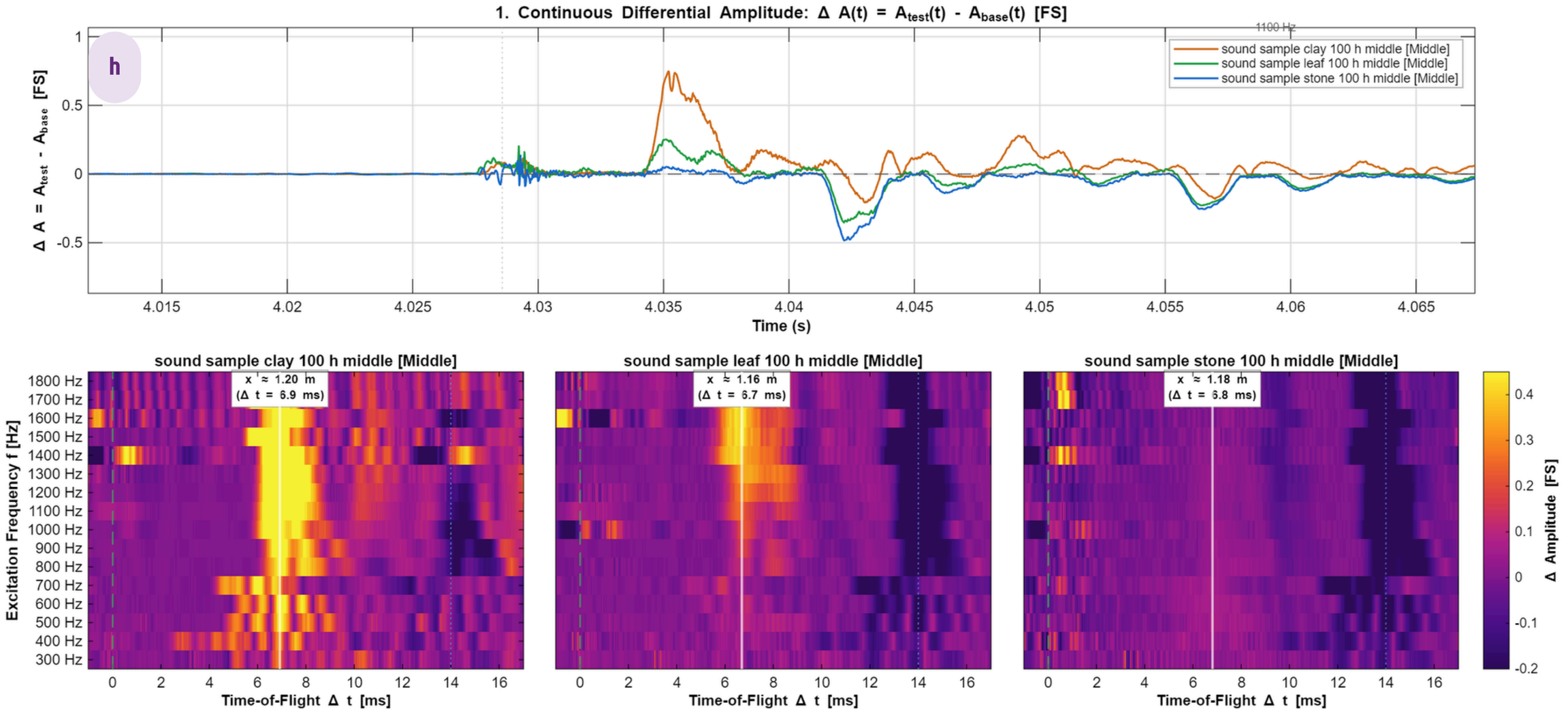
Middle



Experiment 2A Results - Clay, Leaf, Stone Frequency Profile



Experiment 2B Results – Clay, Leaf, Stone Spectrogram Profile



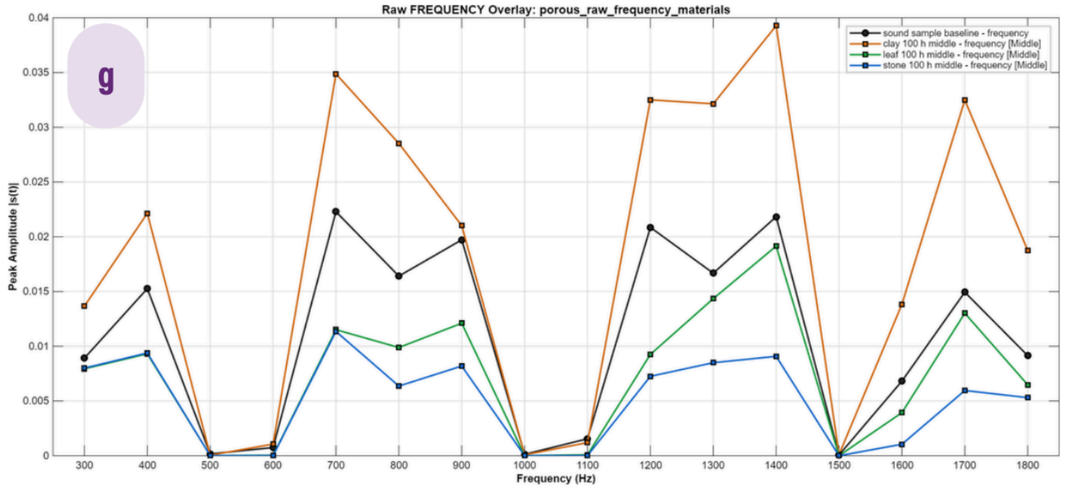
Experiment 2B Results - Conclusion & Discussion

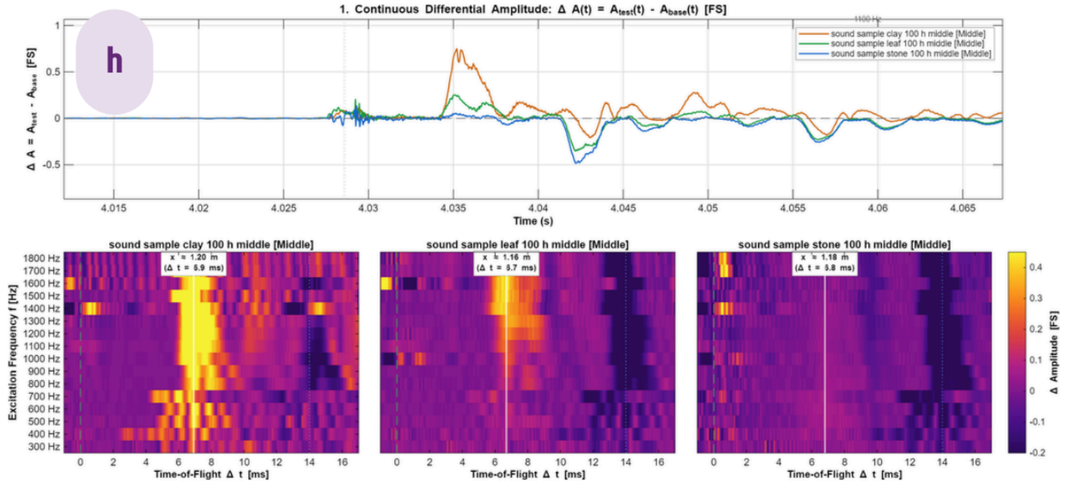
3. FREQUENCY SPECTRUM & STFT ENERGY

Transmission Loss: Delta_dB = 20 * log10(|Y_test| / |Y_base|)
Comb Interference Notch: x = v / (2 * Delta_f) ==> Delta_f approx 173 Hz for x = 1.0 m

Spectral & Time-Frequency Findings (Plots g & h):

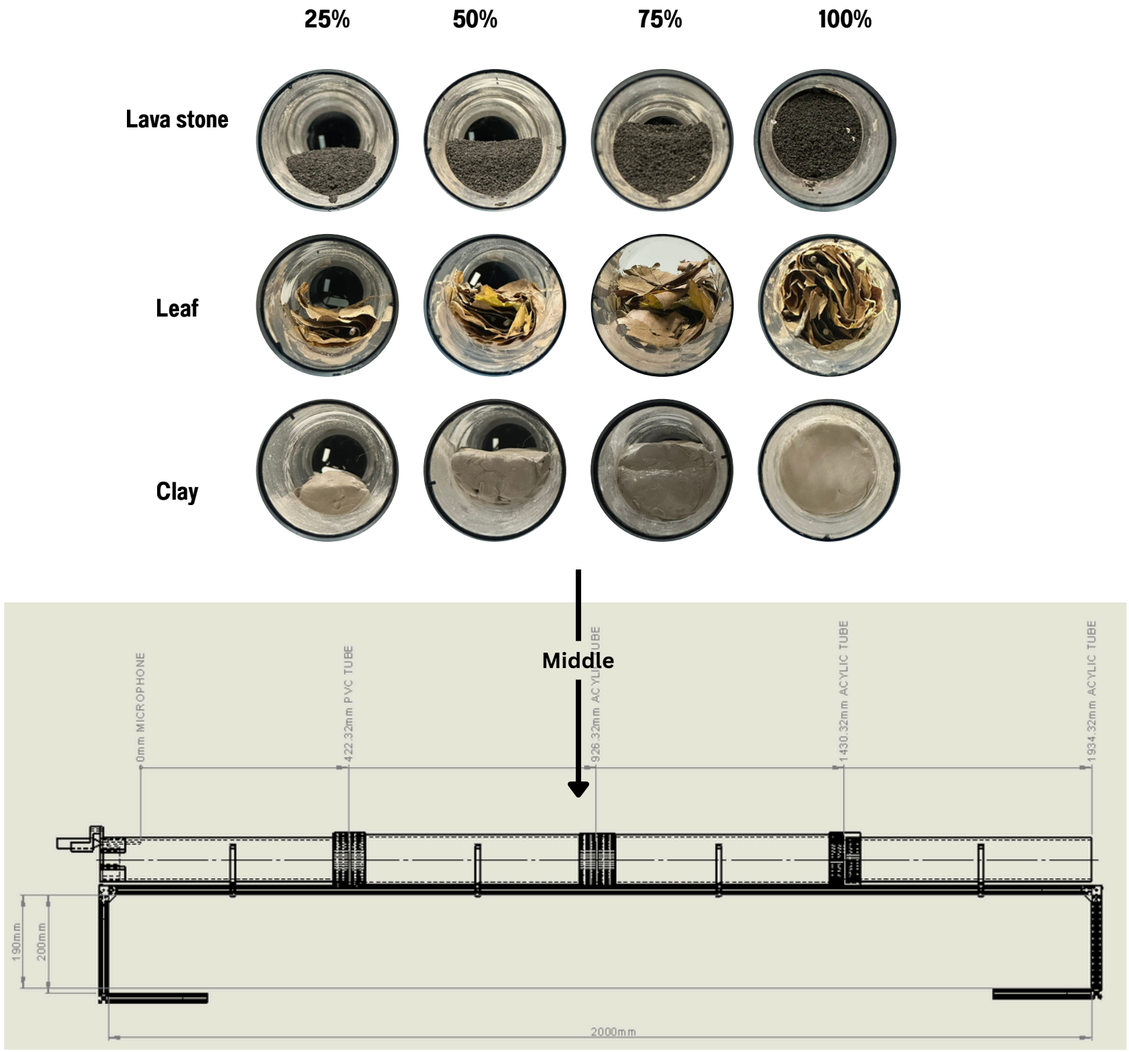
* **Clay (Broadband Specular Reflection):**
  Maintains massive reflection amplitude across the entire spectrum (300 to 1800 Hz) with deep standing wave notches.
* **Leaf (Severe High-Frequency Roll-Off):**
  Frequencies above 1200 Hz drop sharply (S < -10 dB/kHz) due to viscous drag in pore channels. High frequencies absorbed first!
* **Comb-Filter Interference Alignment:**
  Destructive cancellation valleys at 500 Hz, 1000 Hz, 1500 Hz verify quarter-wave acoustic blockage at x = 1.0 m.
* **Spectrogram Energy Ridge (Plot h):**
  Clay exhibits continuous bright yellow energy ridge; Leaf energy decays rapidly into purple silence in higher bands.



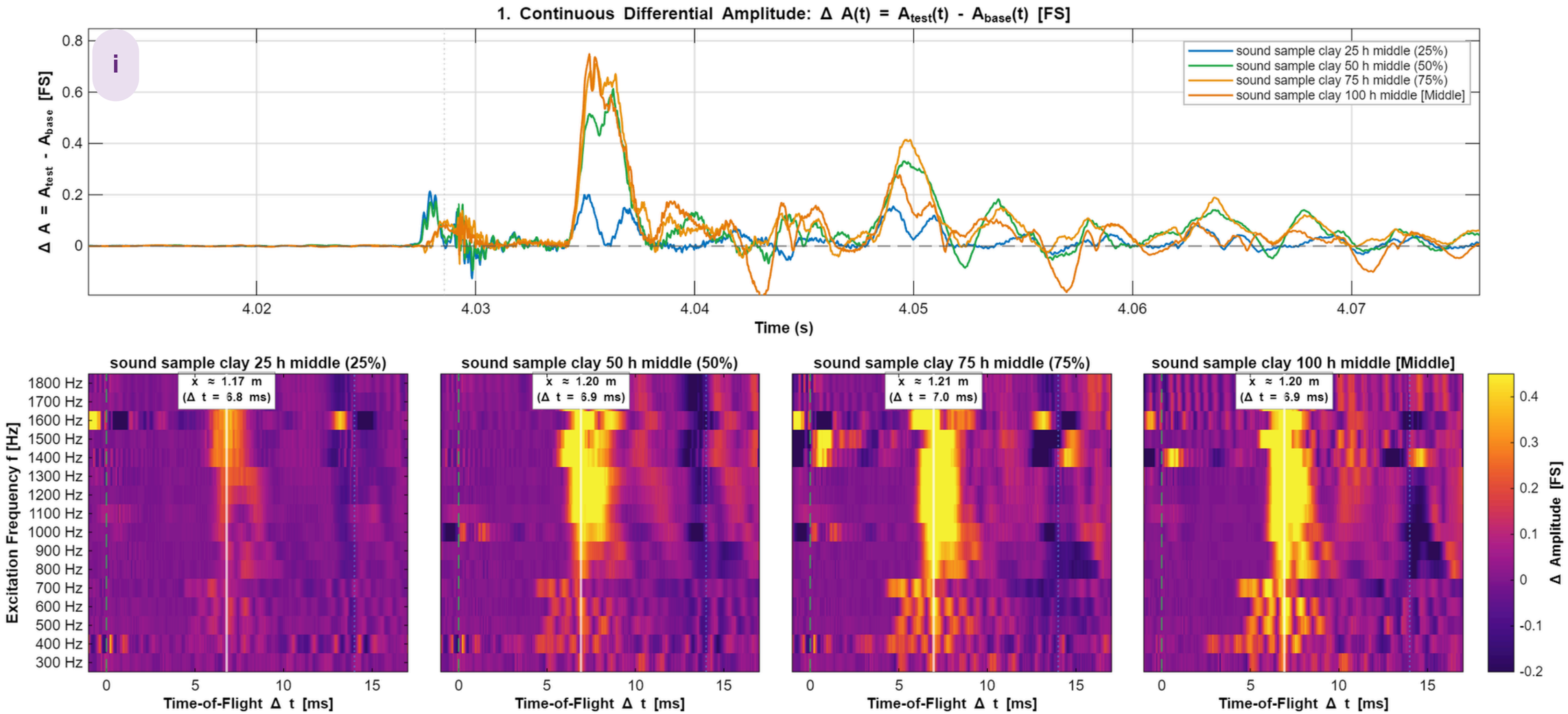


Experiment 3A
Area Blockage

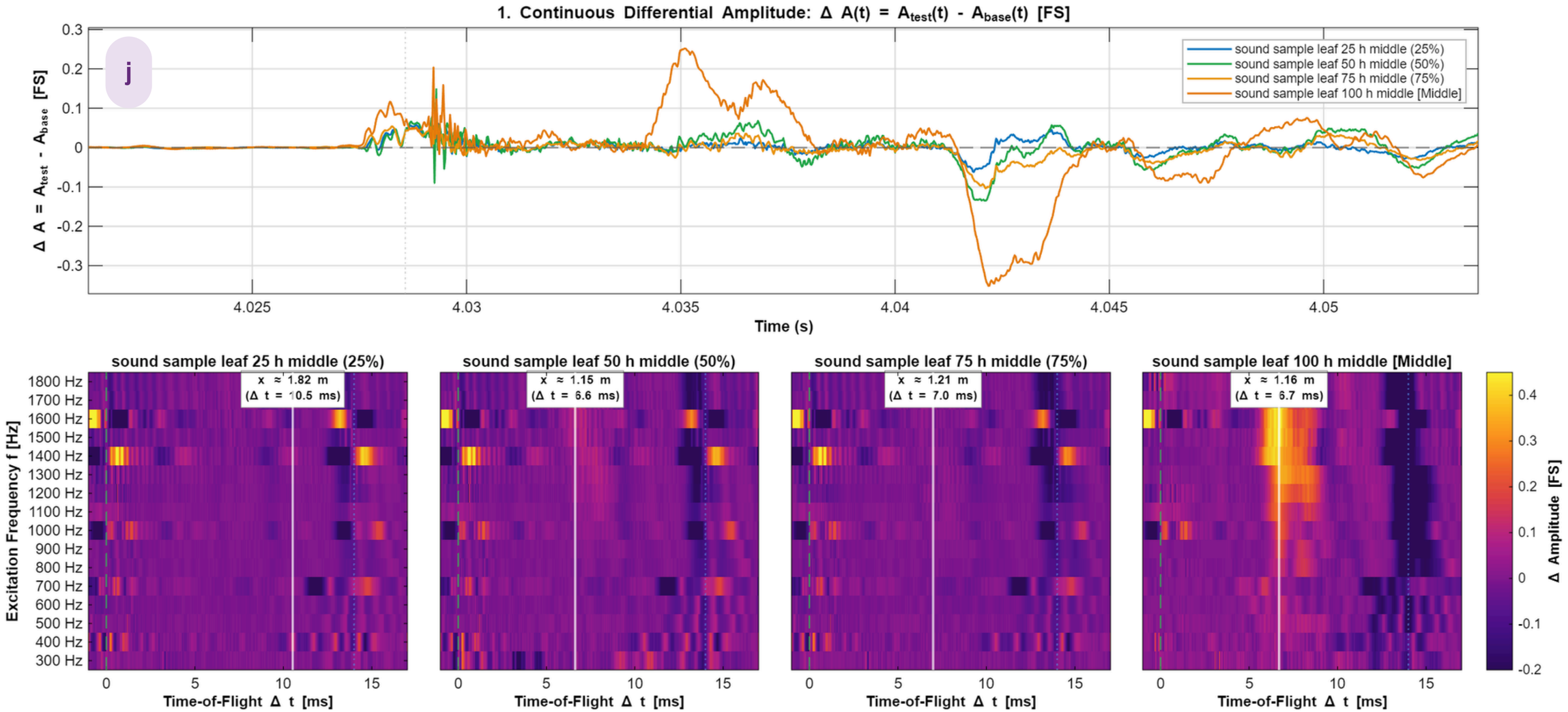
Experiment 3A Procedure



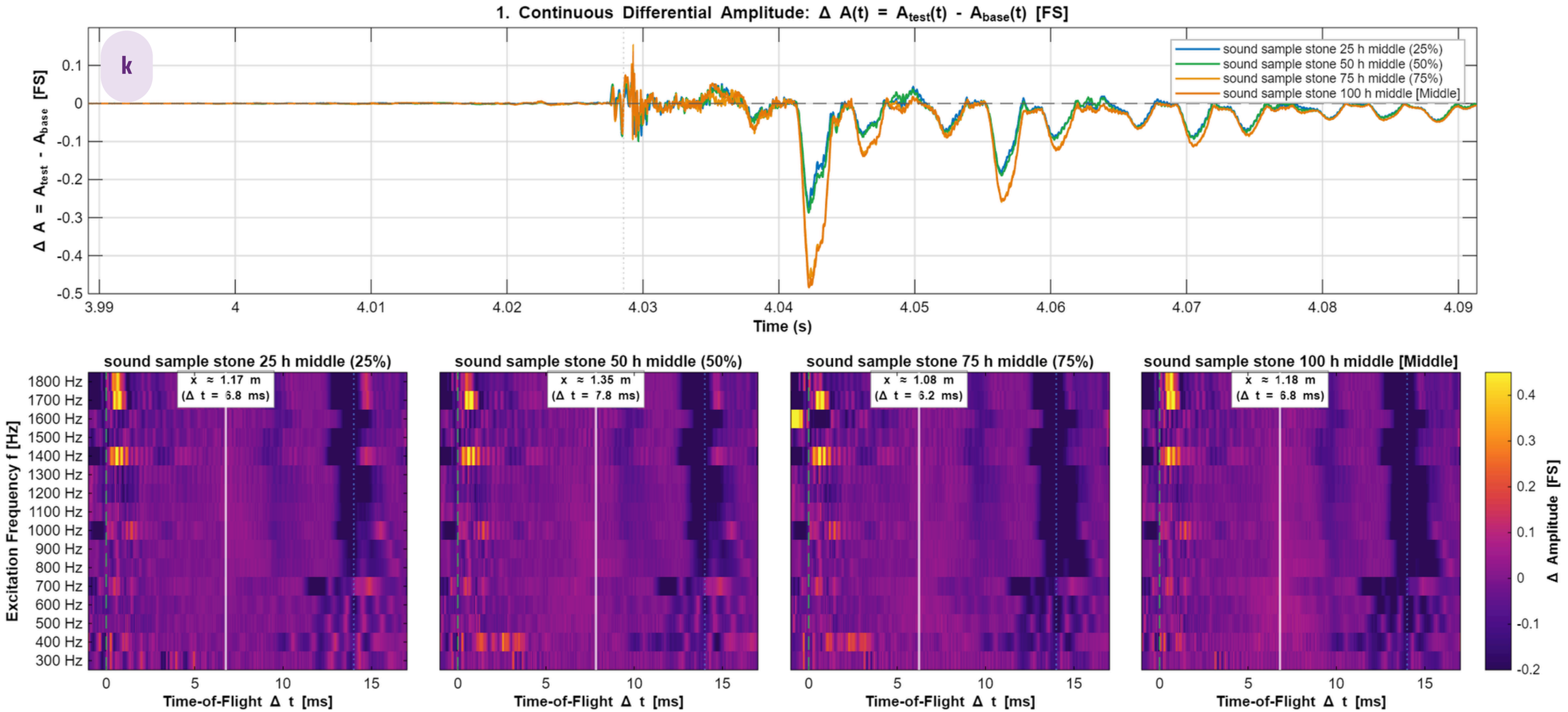
Experiment 3A Results - Clay



Experiment 3A Results



Experiment 3A Results



Experiment 3A Results - Conclusion & Discussion

4. AREA BLOCKAGE ASSESSMENT

Transmission Line Model: alpha = 2R / (1 + R) <===> R = alpha / (2 - alpha)
Area Constriction: S_1 = S_0 * (1 - alpha) ==> Impedance Jump: Z_1 = Z_0 / (1 - alpha)

Blockage Progression Analysis (Plots i, j, k):

* **Monotonic Reflection Growth:**
  As blockage area rises (25% -> 50% -> 75% -> 100%), reflection echo magnitude Delta A escalates monotonically.
* **Clay Scaling (Plot i):**
  Follows the theoretical transmission line curve precisely, reaching R = 1.00 at full cross-section blockage (100%).
* **Leaf Scaling (Plot j):**
  Reflection increases progressively with area but reaches a lower plateau due to internal porous sound absorption.
* **Lava Stone Scaling (Plot k):**
  Causes massive through-pipe transmission loss while backscatter remains diffused by jagged surface facets.
* **Decoupled 2-Stage Diagnostic Suite:**
  1. Where is it? ==> Time-of-Flight delay Delta_t
  2. How much is blocked? ==> Reflection coefficient alpha = 2R / (1 + R)